

```
#include <stdio.h>

void solidSquare()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

void solidRectangle()
{
    int m, n, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);

    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

void rowNumberPattern()
{
    int m, n, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);

    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
        {
            printf("%d ", i);
        }
    }
}
```

```

        printf("\n");
    }
}

void columnNumberPattern()
{
    int m, n, i, j;

    printf("Enter number of rows and columns: ");
    scanf("%d %d", &m, &n);

    for (i = 1; i <= m; i++)
    {
        for (j = 1; j <= n; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }
}

int main()
{
    int choice;

    printf("----- PATTERN MENU ----- \n");
    printf("1. Solid Square Pattern \n");
    printf("2. Solid Rectangular Pattern \n");
    printf("3. Row-Number Pattern \n");
    printf("4. Column-Number Pattern \n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            solidSquare();
            break;

        case 2:
            solidRectangle();
            break;

        case 3:
            rowNumberPattern();
            break;

        case 4:

```

```
        columnNumberPattern();  
        break;  
  
    default:  
        printf("Invalid choice\n");  
}  
  
return 0;  
}
```

Algorithms

1. Solid Square Pattern

1. **Start.**
2. **Input n .**
3. **Set $i = 1$.**
4. **Repeat while $i \leq n$.**
5. **Use an inner loop from $j = 1$ to n .**
6. **Print $*$ in each iteration.**
7. **After the inner loop, move to the next line.**
8. **Stop after n rows.**

2. Solid Rectangular Pattern

1. **Start.**
2. **Input number of rows m and columns n .**
3. **Run the outer loop from 1 to m .**
4. **Run the inner loop from 1 to n .**
5. **Print $*$ in each inner-loop iteration.**
6. **Move to the next line after each row.**
7. **Stop.**

3. Row-Number Pattern

1. Start.
2. Input m rows and n columns.
3. Run the outer loop using i from 1 to m .
4. Run the inner loop using j from 1 to n .
5. Print the value of i .
6. Move to the next line after each row.
7. Stop.

4. Column-Number Pattern

1. Start.
2. Input m rows and n columns.
3. Run the outer loop using i from 1 to m .
4. Run the inner loop using j from 1 to n .
5. Print the value of j .
6. Move to the next line after each row.
7. Stop.

Test-Case Table

Pattern	Input	Expected Output	Result
Solid Square	Choice = 1, $n = 4$	4×4 solid * pattern	Pass
Solid	Choice = 2, $m = 3$, $n = 5$	3×5 solid * pattern	Pass
Row Number	Choice = 3, $m = 4$, $n = 4$	Rows containing 1 1 1 1 through 4 4 4 4	Pass
Column	Choice = 4, $m = 4$, $n = 4$	Each row contains 1 2 3 4	Pass

```
dhruvpandey@DHruVs-MacBook-Air ~ % clang day23.c -o day23
dhruvpandey@DHruVs-MacBook-Air ~ % ./day23
```

```
----- PATTERN MENU -----
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
```

```
Enter your choice: 1
Enter number of rows: 4
```

```
*****
*****
*****
*****
```

```
dhruvpandey@DHruVs-MacBook-Air ~ % ./day23
```

```
----- PATTERN MENU -----
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
```

```
Enter your choice: 2
Enter number of rows and columns: m
```

```
dhruvpandey@DHruVs-MacBook-Air ~ % ./day23
```

```
----- PATTERN MENU -----
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
```

```
Enter your choice: 2
Enter number of rows and columns: 3
```

```
5
*****
*****
*****
```

```
dhruvpandey@DHruVs-MacBook-Air ~ % ./day23
```

```
----- PATTERN MENU -----
1. Solid Square Pattern
2. Solid Rectangular Pattern
3. Row-Number Pattern
4. Column-Number Pattern
```

```
Enter your choice: 4
Enter number of rows and columns: 4
```

```
4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

```
dhruvpandey@DHruVs-MacBook-Air ~ % █
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